

Network Traffic Analysis using Wireshark

Introduction

Network traffic analysis is an important part of Cybersecurity that helps professionals or network admins to understand how devices communicate over the network. This also helps them to predict upcoming threats that might cause networks to stall or identify potential security issues. By capturing the network through Wireshark I was able to observe various protocols, monitor protocols, etc.

Objective

1. Understand the fundamentals of network traffic analysis.
2. Capture live network packets using Wireshark.
3. Analyze common networking protocols.
4. Examine TCP/IP communication.
5. Observe DNS queries and responses.
6. Monitor ICMP traffic.

Lab Environment

1. Operating System - Kali Linux
2. Tools - Wireshark
3. Network Interface - Ethernet
4. Browser - Firefox
5. Protocols - DNS, ICMP, TCP, UDP, HTTPS, HTTP, ARP

Methodology

1. Packet Capture

1. Launched Wireshark and selected the active network interface.
2. Started a live packet capture session.
3. Generated network traffic by browsing websites and executing network commands.
4. Captured packets for analysis.

The image shows the Wireshark network protocol analyzer interface. The top pane displays a list of captured packets with columns for No., Time, Source, Destination, Protocol, and Length. The middle pane shows the details of the selected packet (No. 470), including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Transport Layer Security. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
2	0.513843971	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
3	1.024541892	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
4	1.536434386	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
5	2.047981456	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
6	2.456809179	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
7	2.970810999	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
8	3.481852837	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
9	4.302289556	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
10	4.507623524	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
11	5.324852378	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
12	5.428537974	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
13	6.360395133	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
14	6.451558099	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
15	7.373562087	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
16	7.475552592	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
17	8.294729455	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
18	8.499821580	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
19	9.704380218	192.168.1.37	239.255.255.250	SSDP	470	NOTIFY * HTTP/1.1
20	9.319432828	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
21	9.523681978	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
22	9.582823770	192.168.1.37	239.255.255.250	SSDP	485	NOTIFY * HTTP/1.1
23	10.073850520	192.168.1.33	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
24	10.287369882	192.168.1.37	239.255.255.250	SSDP	550	NOTIFY * HTTP/1.1
25	10.342837922	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
26	10.445298421	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
27	11.048728740	192.168.1.37	239.255.255.250	SSDP	540	NOTIFY * HTTP/1.1
28	11.119575199	192.168.1.37	239.255.255.250	SSDP	542	NOTIFY * HTTP/1.1
29	11.250601351	192.168.1.37	239.255.255.250	SSDP	528	NOTIFY * HTTP/1.1
30	11.367318193	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
31	11.468759724	GXIndia_88:77:60	Broadcast	ARP	60	Who has 192.168.1.40? Tell 192.168.1.1
32	11.733339323	192.168.1.37	239.255.255.250	SSDP	470	NOTIFY * HTTP/1.1

2. DNS Analysis

1. Applied DNS Filter to observe Domain Name Resolution

The image shows the Wireshark network protocol analyzer interface with the filter 'dns' applied. The top pane displays a list of captured DNS packets. The middle pane shows the details of the selected packet (No. 73), including Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (response). The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
62	20.671126990	192.168.1.49	192.168.1.1	DNS	75	Standard query 0x3db5 A ads.mozilla.org
63	20.671562143	192.168.1.49	192.168.1.1	DNS	97	Standard query 0xe374 A firefox.settings.services.mozilla.com
64	20.671710979	192.168.1.49	192.168.1.1	DNS	97	Standard query 0xb772 AAAA firefox.settings.services.mozilla.com
65	20.671798716	192.168.1.49	192.168.1.1	DNS	75	Standard query 0x0849 AAAA ads.mozilla.org
66	20.674095475	192.168.1.49	192.168.1.1	DNS	95	Standard query 0x04c9 A content-signature-2.cdn.mozilla.net
67	20.675453977	192.168.1.49	192.168.1.1	DNS	95	Standard query 0x3fcb AAAA content-signature-2.cdn.mozilla.net
68	20.677744437	192.168.1.1	192.168.1.49	DNS	127	Standard query response 0x3db5 A ads.mozilla.org CNAME mozilla.map.fastly.net A 151.101.209.91
69	20.677744757	192.168.1.1	192.168.1.49	DNS	149	Standard query response 0xe374 A firefox.settings.services.mozilla.com CNAME mozilla.map.fastly.net A 151.101.209.91
70	20.681742530	192.168.1.1	192.168.1.49	DNS	161	Standard query response 0xb772 AAAA firefox.settings.services.mozilla.com CNAME mozilla.map.fastly.net AAAA 2a04:4e42:31::347
71	20.681743037	192.168.1.1	192.168.1.49	DNS	139	Standard query response 0x0849 AAAA ads.mozilla.org CNAME mozilla.map.fastly.net AAAA 2a04:4e42:31::347
72	20.681743885	192.168.1.1	192.168.1.49	DNS	111	Standard query response 0x04c9 A content-signature-2.cdn.mozilla.net A 34.160.144.191
73	20.681743915	192.168.1.1	192.168.1.49	DNS	123	Standard query response 0x3fcb AAAA content-signature-2.cdn.mozilla.net A 34.160.144.191
229	22.190798173	192.168.1.49	192.168.1.1	DNS	75	Standard query 0x876e A www.youtube.com
230	22.198892505	192.168.1.49	192.168.1.1	DNS	75	Standard query 0xb595 HTTPS www.youtube.com
231	22.198414548	192.168.1.49	192.168.1.1	DNS	75	Standard query 0xd852 AAAA www.youtube.com
232	22.205092137	192.168.1.1	192.168.1.49	DNS	495	Standard query response 0xb595 HTTPS www.youtube.com CNAME youtube-ui.l.google.com HTTPS A 142.250.67.14 A 142.250.70.46 A 142.250.70.110 A 142.250.70.118 A 142.250.70.119 A 142.250.70.120 A 142.250.70.121 A 142.250.70.122 A 142.250.70.123 A 142.250.70.124 A 142.250.70.125 A 142.250.70.126 A 142.250.70.127 A 142.250.70.128 A 142.250.70.129 A 142.250.70.130 A 142.250.70.131 A 142.250.70.132 A 142.250.70.133 A 142.250.70.134 A 142.250.70.135 A 142.250.70.136 A 142.250.70.137 A 142.250.70.138 A 142.250.70.139 A 142.250.70.140 A 142.250.70.141 A 142.250.70.142 A 142.250.70.143 A 142.250.70.144 A 142.250.70.145 A 142.250.70.146 A 142.250.70.147 A 142.250.70.148 A 142.250.70.149 A 142.250.70.150 A 142.250.70.151 A 142.250.70.152 A 142.250.70.153 A 142.250.70.154 A 142.250.70.155 A 142.250.70.156 A 142.250.70.157 A 142.250.70.158 A 142.250.70.159 A 142.250.70.160 A 142.250.70.161 A 142.250.70.162 A 142.250.70.163 A 142.250.70.164 A 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142.250.70.399 A 142.250.70.400 A 142.250.70.401 A 142.250.70.402 A 142.250.70.403 A 142.250.70.404 A 142.250.70.405 A 142.250.70.406 A 142.250.70.407 A 142.250.70.408 A 142.250.70.409 A 142.250.70.410 A 142.250.70.411 A 142.250.70.412 A 142.250.70.413 A 142.250.70.414 A 142.250.70.415 A 142.250.70.416 A 142.250.70.417 A 142.250.70.418 A 142.250.70.419 A 142.250.70.420 A 142.250.70.421 A 142.250.70.422 A 142.250.70.423 A 142.250.70.424 A 142.250.70.425 A 142.250.70.426 A 142.250.70.427 A 142.250.70.428 A 142.250.70.429 A 142.250.70.430 A 142.250.70.431 A 142.250.70.432 A 142.250.70.433 A 142.250.70.434 A 142.250.70.435 A 142.250.70.436 A 142.250.70.437 A 142.250.70.438 A 142.250.70.439 A 142.250.70.440 A 142.250.70.441 A 142.250.70.442 A 142.250.70.443 A 142.250.70.444 A 142.250.70.445 A 142.250.70.446 A 142.250.70.447 A 142.250.70.448 A 142.250.70.449 A 142.250.70.450 A 142.250.70.451 A 142.250.70.452 A 142.250.70.453 A 142.250.70.454 A 142.250.70.455 A 142.250.70.456 A 142.250.70.457 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142.250.70.516 A 142.250.70.517 A 142.250.70.518 A 142.250.70.519 A 142.250.70.520 A 142.250.70.521 A 142.250.70.522 A 142.250.70.523 A 142.250.70.524 A 142.250.70.525 A 142.250.70.526 A 142.250.70.527 A 142.250.70.528 A 142.250.70.529 A 142.250.70.530 A 142.250.70.531 A 142.250.70.532 A 142.250.70.533 A 142.250.70.534 A 142.250.70.535 A 142.250.70.536 A 142.250.70.537 A 142.250.70.538 A 142.250.70.539 A 142.250.70.540 A 142.250.70.541 A 142.250.70.542 A 142.250.70.543 A 142.250.70.544 A 142.250.70.545 A 142.250.70.546 A 142.250.70.547 A 142.250.70.548 A 142.250.70.549 A 142.250.70.550 A 142.250.70.551 A 142.250.70.552 A 142.250.70.553 A 142.250.70

3. TCP Communication Analysis

1. Filtered TCP packets
2. Examined the three way handshake (SYN, SYN-ACK, ACK)

tcp.flags.syn && tcp.flags.ack					
No.	Time	Source	Destination	Protocol	Length Info
74	20.694713961	192.168.1.49	151.101.209.91	TCP	74 48248 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=360739097 TSecr=0 WS=512
75	20.695060243	192.168.1.49	34.160.144.191	TCP	74 47792 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=749357379 TSecr=0 WS=512
76	20.696532492	192.168.1.49	34.160.144.191	TCP	74 47896 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2717031088 TSecr=0 WS=512
77	20.697076746	192.168.1.49	34.160.144.191	TCP	74 47810 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=143782378 TSecr=0 WS=512
78	20.700999633	151.101.209.91	192.168.1.49	TCP	74 443 → 48246 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1452 SACK_PERM TSval=2604704736 TSecr=3007390997 WS=512
79	20.700999163	34.160.144.191	192.168.1.49	TCP	74 443 → 47792 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSval=1874011252 TSecr=749357379 WS=256
80	20.700999192	34.160.144.191	192.168.1.49	TCP	74 443 → 47896 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSval=992831859 TSecr=2717031088 WS=256
81	20.701754428	192.168.1.49	151.101.209.91	TCP	66 48246 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=3607390904 TSecr=2604704736
82	20.702117820	192.168.1.49	34.160.144.191	TCP	66 47792 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=749357385 TSecr=1874011252
83	20.702311898	192.168.1.49	34.160.144.191	TCP	66 47896 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2717031014 TSecr=992831859

tcp					
No.	Time	Source	Destination	Protocol	Length Info
74	20.694713961	192.168.1.49	151.101.209.91	TCP	74 48248 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=360739097 TSecr=0 WS=512
75	20.695060243	192.168.1.49	34.160.144.191	TCP	74 47792 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=749357379 TSecr=0 WS=512
76	20.696532492	192.168.1.49	34.160.144.191	TCP	74 47896 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2717031088 TSecr=0 WS=512
77	20.697076746	192.168.1.49	34.160.144.191	TCP	74 47810 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=143782378 TSecr=0 WS=512
78	20.700999633	151.101.209.91	192.168.1.49	TCP	74 443 → 48246 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1452 SACK_PERM TSval=2604704736 TSecr=3007390997 WS=512
79	20.700999163	34.160.144.191	192.168.1.49	TCP	74 443 → 47792 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSval=1874011252 TSecr=749357379 WS=256
80	20.700999192	34.160.144.191	192.168.1.49	TCP	74 443 → 47896 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSval=992831859 TSecr=2717031088 WS=256
81	20.701754428	192.168.1.49	151.101.209.91	TCP	66 48246 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=3607390904 TSecr=2604704736
82	20.702117820	192.168.1.49	34.160.144.191	TCP	66 47792 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=749357385 TSecr=1874011252
83	20.702311898	192.168.1.49	34.160.144.191	TCP	66 47896 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=2717031014 TSecr=992831859
84	20.705494264	34.160.144.191	192.168.1.49	TCP	74 443 → 47810 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM TSval=1166758726 TSecr=143782378 WS=256
85	20.705492330	192.168.1.49	34.160.144.191	TCP	66 47810 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=143702307 TSecr=1166750726
86	20.713835538	192.168.1.49	151.101.209.91	TLSv1.2	288 Client Hello (SN:firefox.settings.services.mozilla.com)
87	20.715264496	192.168.1.49	34.160.144.191	TLSv1.2	286 Client Hello (SN:content-signature-2.cdn.mozilla.net)
88	20.715767836	192.168.1.49	34.160.144.191	TLSv1.2	286 Client Hello (SN:content-signature-2.cdn.mozilla.net)
89	20.716034946	192.168.1.49	34.160.144.191	TLSv1.2	286 Client Hello (SN:content-signature-2.cdn.mozilla.net)
90	20.720187425	151.101.209.91	192.168.1.49	TCP	66 443 → 48246 [ACK] Seq=1 Ack=223 Win=145920 Len=0 TSval=2604704754 TSecr=3007390917
91	20.720187739	151.101.209.91	192.168.1.49	TLSv1.2	4386 Server Hello, Certificate
92	20.720471075	192.168.1.49	151.101.209.91	TCP	66 48246 → 443 [ACK] Seq=223 Ack=4321 Win=73216 Len=0 TSval=3007390923 TSecr=2604704755
93	20.723844148	151.101.209.91	192.168.1.49	TLSv1.2	293 Server Key Exchange, Server Hello Done
94	20.723844649	34.160.144.191	192.168.1.49	TCP	66 443 → 47810 [ACK] Seq=1 Ack=221 Win=268800 Len=0 TSval=1166758743 TSecr=143782397
95	20.723844693	34.160.144.191	192.168.1.49	TCP	66 443 → 47792 [ACK] Seq=1 Ack=221 Win=268800 Len=0 TSval=1874011273 TSecr=749357399
96	20.723844740	34.160.144.191	192.168.1.49	TCP	66 443 → 47896 [ACK] Seq=1 Ack=221 Win=268800 Len=0 TSval=992831879 TSecr=2717031027
97	20.723885110	192.168.1.49	151.101.209.91	TCP	66 48246 → 443 [ACK] Seq=223 Ack=4548 Win=76288 Len=0 TSval=3007390927 TSecr=2604704755
98	20.754323230	192.168.1.49	151.101.209.91	TLSv1.2	159 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
99	20.759404369	192.168.1.49	151.101.209.91	TCP	74 48248 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=3226170273 TSecr=0 WS=512
100	20.759506931	151.101.209.91	192.168.1.49	TCP	66 443 → 48246 [ACK] Seq=4548 Ack=316 Win=145920 Len=0 TSval=2604704795 TSecr=3007390957
101	20.759507263	151.101.209.91	192.168.1.49	TLSv1.2	324 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
102	20.760446937	151.101.209.91	192.168.1.49	TCP	74 48248 → 443 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1452 SACK_PERM TSval=1521067893 TSecr=3226170273 WS=512
103	20.764893591	192.168.1.49	151.101.209.91	TCP	66 48248 → 443 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=3226170279 TSecr=1521067893
104	20.770888004	192.168.1.49	151.101.209.91	TLSv1.3	1906 Client Hello (SN:ads.mozilla.org)
105	20.770899573	151.101.209.91	192.168.1.49	TCP	66 443 → 48248 [ACK] Seq=1 Ack=1441 Win=143360 Len=0 TSval=1521067905 TSecr=3226170284

Frame 74: Packet, 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eth0, id 0

Ethernet II, Src: PCSysentec 78:ce:b7 (08:00:27:78:ce:b7), Dst: GxIndia 88:77:60 (20:0c:86:88:77:60)

Internet Protocol Version 4, Src: 192.168.1.49, Dst: 151.101.209.91

Transmission Control Protocol, Src Port: 48248, Dst Port: 443, Seq: 0, Len: 0

0000 20 0c 86 88 77 60 08 00 27 78 ce b7 08 00 45 00 w'x'E

0010 00 3c 25 ec 40 00 40 00 ea 35 c0 a8 01 31 97 05 j:00 d:1e

0020 01 5b d7 0c 00 5b 21 df a9 68 cd 3b 0d d7 80 18 [P:]

0030 00 7e 2b f7 00 00 01 01 00 0a ec 64 96 cd be 29 +-----d-----

0040 42 08 47 45 54 20 2f 73 75 63 03 05 73 2e 74 B GET /s. success.t

0050 78 74 3f 69 70 76 34 20 48 54 54 50 2f 31 2e 31 xt?ipv4 HTTP/1.1

0060 6d 0a 4b 0f 73 74 3a 20 64 05 74 05 63 74 70 6f Host: detectpro

0070 72 74 61 0c 2e 66 69 72 65 66 6f 78 2e 63 6f 6d rta.fir efox.com

0080 6d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f -A gent: Mo

Hypertext Transfer Protocol: Protocol

Packets: 27645 - Displayed: 15 (0.1%) - Dropped: 0 (0.0%)

4. HTTP /HTTPS Packet Analysis

1. Inspected the HTTP and HTTPS packets.
2. Compared the HTTP (unencrypted) and HTTPS (Encrypted) communication.

http					
No.	Time	Source	Destination	Protocol	Length Info
2292	33.287841358	192.168.1.49	151.101.209.91	HTTP	376 GET /success.txt?ip=1 HTTP/1.1
2295	33.296622230	151.101.209.91	192.168.1.49	HTTP	442 HTTP/1.1 200 OK (text/plain)
4911	88.845238104	192.168.1.49	34.223.124.45	HTTP	398 GET / HTTP/1.1
4118	89.845238104	192.168.1.49	34.223.124.45	HTTP	418 [TCP previous segment not captured] Continuation
12592	116.001757582	192.168.1.49	142.250.206.131	OCSP	493 Request
12643	116.074914900	142.250.206.131	192.168.1.49	OCSP	1168 Response
12880	117.134407832	192.168.1.49	142.250.206.131	OCSP	493 Request
12934	117.320087830	142.250.206.131	192.168.1.49	OCSP	1168 Response
20834	131.096268837	192.168.1.49	146.190.62.39	HTTP	351 GET / HTTP/1.1
20835	131.122601770	146.190.62.39	192.168.1.49	HTTP	424 HTTP/1.0 200 OK (text/html)
20858	131.380402913	146.190.62.39	192.168.1.49	HTTP	2822 [TCP Out-of-Order] HTTP/1.1 200 OK (text/html)
20870	131.458118556	192.168.1.49	146.190.62.39	HTTP	519 GET / HTTP/1.1
20871	131.465091442	146.190.62.39	192.168.1.49	HTTP	424 HTTP/1.0 200 OK (text/html)
20883	132.112662931	192.168.1.49	154.18.50.125	HTTP	432 GET /webadmin/deny/ HTTP/1.1
20885	132.123708181	154.18.50.125	192.168.1.49	HTTP	1204 HTTP/1.1 200 OK (text/html)

Frame 2292: Packet, 376 bytes on wire (3008 bits), 376 bytes captured (3008 bits) on interface eth0, id 0

Ethernet II, Src: PCSysentec 78:ce:b7 (08:00:27:78:ce:b7), Dst: GxIndia 88:77:60 (20:0c:86:88:77:60)

Internet Protocol Version 4, Src: 192.168.1.49, Dst: 151.101.209.91

Transmission Control Protocol, Src Port: 55952, Dst Port: 80, Seq: 1, Ack: 1, Len: 310

Hypertext Transfer Protocol

0000 20 0c 86 88 77 60 08 00 27 78 ce b7 08 00 45 00 w'x'E

0010 01 5b d7 0c 00 5b 21 df a9 68 cd 3b 0d d7 80 18 j:00 d:1e

0020 01 5b d7 0c 00 5b 21 df a9 68 cd 3b 0d d7 80 18 [P:]

0030 00 7e 2b f7 00 00 01 01 00 0a ec 64 96 cd be 29 +-----d-----

0040 42 08 47 45 54 20 2f 73 75 63 03 05 73 2e 74 B GET /s. success.t

0050 78 74 3f 69 70 76 34 20 48 54 54 50 2f 31 2e 31 xt?ipv4 HTTP/1.1

0060 6d 0a 4b 0f 73 74 3a 20 64 05 74 05 63 74 70 6f Host: detectpro

0070 72 74 61 0c 2e 66 69 72 65 66 6f 78 2e 63 6f 6d rta.fir efox.com

0080 6d 0a 55 73 65 72 2d 41 67 65 6e 74 3a 20 4d 6f -A gent: Mo

Hypertext Transfer Protocol: Protocol

Packets: 27645 - Displayed: 15 (0.1%) - Dropped: 0 (0.0%)

5. ARP Analysis

1. Checked the broadcasts for MAC address from Router

No.	Time	Source	Destination	Protocol	Length	Info
817	25.497543279	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
883	26.317201659	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
1000	26.521523966	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
1054	27.340759889	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
1059	27.443286267	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
1219	28.467379345	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
1243	28.672442407	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
1383	29.491383536	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
1436	29.697185873	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
1686	30.316264893	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
1670	30.515381804	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
1729	30.720441397	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
1965	31.437806301	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
1959	31.643022238	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
2128	32.461567664	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
2140	32.666276769	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
2327	33.485780910	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
2341	33.690232588	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
2382	34.599002714	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
2389	34.714353598	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
2649	35.430678887	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
2672	35.635524847	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
2928	36.659364217	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.44? Tell 192.168.1.1
3089	37.475043347	GXIndia.88:77:60	Broadcast	ARP	60	Who has 192.168.1.49? Tell 192.168.1.1
Frame 1087: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0						
Ethernet II, Src: PCSSystemtec_78:ce:b7 (08:00:27:78:ce:b7), Dst: GXIndia.88:77:60 (20:0c:86:88:77:60)						
Address Resolution Protocol (reply)						
Hardware type: Ethernet (1)						
Protocol type: IPv4 (0x800)						
Hardware size: 6						
Protocol size: 4						
Opcode: reply (2)						
Sender MAC address: PCSSystemtec_78:ce:b7 (08:00:27:78:ce:b7)						
Sender IP address: 192.168.1.49						
Target MAC address: GXIndia.88:77:60 (20:0c:86:88:77:60)						
Target IP address: 192.168.1.1						

6. ICMP analysis

1. Generated ICMP traffic using the ping command to google.

2. Captured Echo Requests and Echo Replies.

No.	Time	Source	Destination	Protocol	Length	Info
10	10.613634184	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=1/256, ttl=64 (reply in 11)
11	10.624673289	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=1/256, ttl=119 (request in 10)
14	11.613192590	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=2/512, ttl=64 (reply in 15)
15	11.618272297	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=2/512, ttl=119 (request in 14)
17	12.743697739	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=3/768, ttl=64 (reply in 18)
18	12.748091801	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=3/768, ttl=119 (request in 17)
19	13.746444128	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=4/1024, ttl=64 (reply in 20)
20	13.755880245	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=4/1024, ttl=119 (request in 19)
22	14.747474754	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=5/1280, ttl=64 (reply in 23)
23	14.757599931	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=5/1280, ttl=119 (request in 22)
26	15.749026117	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=6/1536, ttl=64 (reply in 27)
27	15.754469179	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=6/1536, ttl=119 (request in 26)
31	16.751795130	192.168.1.49	8.8.8.8	ICMP	98	Echo (ping) request id=0x7846, seq=7/1792, ttl=64 (reply in 32)
32	16.757484338	8.8.8.8	192.168.1.49	ICMP	98	Echo (ping) reply id=0x7846, seq=7/1792, ttl=119 (request in 31)

Frame 10: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_78:ce:b7 (08:00:27:78:ce:b7), Dst: GXIndia.88:77:60 (20:0c:86:88:77:60)

Internet Protocol Version 4, Src: 192.168.1.49, Dst: 8.8.8.8

Internet Control Message Protocol

0000 20 0c 86 88 77 60 08 00 27 78 ce b7 08 00 45 00 ...w...x...E

0010 00 54 01 e1 40 00 40 01 00 df c0 a8 01 31 08 08 ...T...f...1...

0020 08 08 08 08 b0 23 70 46 00 01 a1 bb 43 6a 00 00w...C...

0030 00 00 26 9c 05 00 00 00 00 00 10 11 12 13 14 15 ...&...I"#5%

0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 ...&'()*+,-./012345

0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 ...&'()*+,-./012345

0060 36 37 67

7. HTTP and HTTPS (TLS) Analysis

1. Generated HTTP and HTTPS traffic though firefox browser

No.	Time	Source	Destination	Protocol	Length	Info
86	29.71583553	192.168.1.49	151.101.209.91	TLSv1.2	288	Client Hello (SNI=firefox.settings.services.mozilla.com)
87	29.715264496	192.168.1.49	34.160.144.191	TLSv1.2	286	Client Hello (SNI=content-signature-2.cdn.mozilla.net)
88	29.71567836	192.168.1.49	34.160.144.191	TLSv1.2	286	Client Hello (SNI=content-signature-2.cdn.mozilla.net)
89	29.71603496	192.168.1.49	34.160.144.191	TLSv1.2	286	Client Hello (SNI=content-signature-2.cdn.mozilla.net)
91	29.72618739	151.101.209.91	192.168.1.49	TLSv1.2	4386	Server Hello, Certificate
93	29.72844148	151.101.209.91	192.168.1.49	TLSv1.2	293	Server Key Exchange, Server Hello Done
94	29.73049239	151.101.209.91	192.168.1.49	TLSv1.2	158	Change Cipher Spec, Encrypted Handshake Message
101	29.759597263	151.101.209.91	192.168.1.49	TLSv1.2	324	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
104	29.77088894	192.168.1.49	151.101.209.91	TLSv1.3	1966	Client Hello (SNI=ads.mozilla.org)
187	29.77906156	151.101.209.91	192.168.1.49	TLSv1.3	1946	Server Hello, Change Cipher Spec, Application Data
192	29.78200000	151.101.209.91	192.168.1.49	TLSv1.3	282	[TCP Previous segment not captured] - continuation Data
193	29.787401769	34.160.144.191	192.168.1.49	TLSv1.2	286	[TCP Previous segment not captured] - Ignored Unknown Record
112	28.787402443	34.160.144.191	192.168.1.49	TLSv1.2	3086	Server Hello, Certificate, Server Key Exchange, Server Hello Done
113	28.790714541	34.160.144.191	192.168.1.49	TLSv1.2	4386	Server Hello, Certificate, Server Key Exchange, Server Hello Done
117	28.790711656	192.168.1.49	34.160.144.191	TLSv1.2	159	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
119	28.795474364	34.160.144.191	192.168.1.49	TLSv1.2	377	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
120	28.795474397	34.160.144.191	192.168.1.49	TLSv1.2	135	Application Data
129	28.820450891	192.168.1.49	151.101.209.91	TLSv1.2	165	Application Data
130	28.821419883	192.168.1.49	151.101.209.91	TLSv1.2	261	Application Data
131	28.821957746	192.168.1.49	151.101.209.91	TLSv1.3	138	Change Cipher Spec, Application Data
132	28.824731219	192.168.1.49	151.101.209.91	TLSv1.3	158	Application Data
133	28.824607448	192.168.1.49	151.101.209.91	TLSv1.3	270	Application Data
134	28.824601149	192.168.1.49	151.101.209.91	TLSv1.3	322	Application Data
136	28.834994983	151.101.209.91	192.168.1.49	TLSv1.2	138	Application Data
139	28.837246139	151.101.209.91	192.168.1.49	TLSv1.2	1164	Application Data
141	28.839601461	151.101.209.91	192.168.1.49	TLSv1.3	131	Application Data
148	28.935907285	192.168.1.49	151.101.209.91	TLSv1.2	104	Application Data
150	28.963898965	192.168.1.49	151.101.209.91	TLSv1.2	97	Application Data
151	28.963825639	192.168.1.49	34.160.144.191	TLSv1.2	165	Application Data
152	28.964477993	192.168.1.49	34.160.144.191	TLSv1.2	318	Application Data
153	28.964983211	192.168.1.49	34.160.144.191	TLSv1.2	104	Application Data
155	28.970813875	34.160.144.191	192.168.1.49	TLSv1.2	104	Application Data

• Frame 80: Packet, 288 bytes on wire (2304 bits), 288 bytes captured (2304 bits) on interface eth0, id 0

• Ethernet II, Src: PCSysnetSec, 78:c6:b7:88:20:7f, Dst: Xidma:88:77:60:00 (Len: 0x0c:88:77:60:00)

• Internet Protocol Version 4, Src: 192.168.1.49, Destination: 151.101.209.91

• Transmission Control Protocol, Src Port: 8820, Dst Port: 443, Seq.: 1, Len: 222

• Transport Layer Security

```

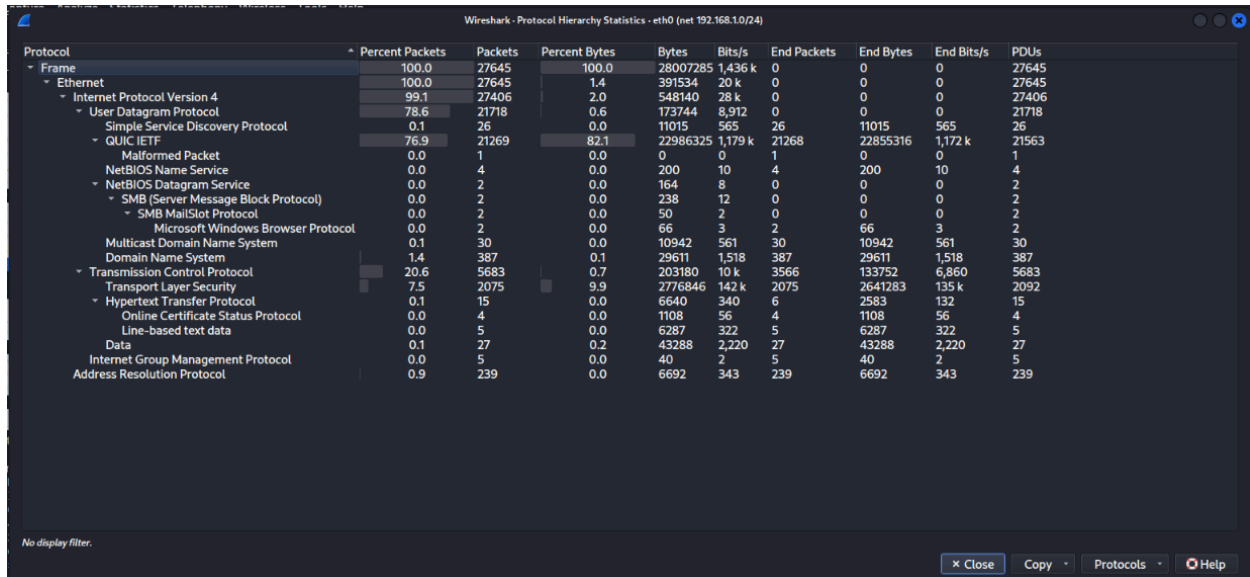
0000  20 0c 86 88 77 60 00 00 27 78 c b 07 00 00 45 00   w' x' E
0010  01 12 25 e5 0e 40 00 40 09 5d 0c 0a 31 01 93 05   S @ ( ) | e
0020  01 5b bc 76 01 b1 cc cc 08 00 63 01 43 00 80 1b   [ v ~ 90
0030  76 b7 2f 00 00 01 01 01 08 a b 41 21 41 9b 40   AIA@
0040  a3 0e 16 03 01 09 01 00 00 05 03 03 ec 0c f      A = b ( l o
0050  01 0c 75 cf 3d 02 90 d9 00 52 2a 64 7b 05 02   A u b h r : b
0060  fa db da 75 3d 5b 05 72 cc f9 7e 94 0e 00 1e   C r f
0070  0c 2b cb 2f cc 0c ac 0c 2c 3b 0c 0a c0 c9      + / . 0
0080  c3 13 00 00 9c 9c 0d 9d 2f 30 35 01 00 00 90   / 5
    
```

No.	Time	Source	Destination	Protocol	Length	Info
2292	33.267841358	192.168.1.49	151.101.209.91	HTTP	376	GET /success.txt?ip=4 HTTP/1.1
2293	33.299622230	151.101.209.91	192.168.1.49	HTTP	442	HTTP/1.1 200 OK (text/plain)
4911	88.845238184	192.168.1.49	34.223.124.45	HTTP	398	GET / HTTP/1.1
4913	89.114914096	34.223.124.45	192.168.1.49	HTTP	891	[TCP Previous segment not captured] Continuation
12592	116.601737582	192.168.1.49	142.250.206.131	OCSP	493	Request
12643	116.674914900	142.250.206.131	192.168.1.49	OCSP	1168	Response
12880	117.134407832	192.168.1.49	142.250.206.131	OCSP	493	Request
12934	117.328087830	142.250.206.131	192.168.1.49	OCSP	1168	Response
20934	131.096268937	192.168.1.49	146.190.62.39	HTTP	351	GET / HTTP/1.1
20935	131.122681770	146.190.62.39	192.168.1.49	HTTP	424	HTTP/1.0 200 OK (text/html)
20958	131.358482313	146.190.62.39	192.168.1.49	HTTP	2822	[TCP Out-Of-Order] HTTP/1.1 200 OK (text/html)
20970	131.458110556	192.168.1.49	146.190.62.39	HTTP	519	GET / HTTP/1.1
20971	131.465091442	146.190.62.39	192.168.1.49	HTTP	424	HTTP/1.0 200 OK (text/html)
20983	132.112662931	192.168.1.49	154.18.50.125	HTTP	432	GET /webadmin/deny/ HTTP/1.1
20985	132.123708181	154.18.50.125	192.168.1.49	HTTP	1284	HTTP/1.1 200 OK (text/html)

Frame 2292: Packet, 376 bytes on wire (3008 bits), 376 bytes captured (3008 bits) on interface eth0, id 0000	20 0c 86 88 77 66 08 00	27 78 ce b7 08 00 45 00	...w...x...E
Ethernet II, Src: PC5Sysmtec.78:ce:b7 (08:00:27:78:ce:b7), Dst: GxIndia.88:77:66 (20:0c:86:88:77:66)	01 01 6a 0e 90 40 08 40 06	00 64 c0 a8 01 31 97 05	...j...@...d...e
Internet Protocol Version 4, Src: 192.168.1.49, Dst: 151.101.209.91	0020	d1 5b d7 0c 00 50 21 df	...[...P...]
Transmission Control Protocol, Src Port: 55952, Dst Port: 80, Seq: 1, Ack: 1, Len: 310	0030	08 7e 2b 17 00 00 01 01	...8...a...c...6...9...d...)
Hypertext Transfer Protocol	0040	42 d8 47 45 54 28 2f 73	...B...G.../...s...u...c...c...e...s...s...t...)
	0050	78 74 3f 69 70 76 34 20	...xt7ipv4 HTTP/1.1
	0060	8d 0a 48 6f 73 74 3a 20	...Host: detectpo
	0070	72 74 61 6c 2e 06 09 72	...rtal.fir efox.com
	0080	6d 0a 55 73 65 72 2d 4f	...User-A gent: Mo

8. Protocol Hierarchy Review

1. In the Statistics tab → Protocol hierarchy checked the the protocol distribution and packet count.



Wireshark - Protocol Hierarchy Statistics - eth0 (net 192.168.1.0/24)

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	27645	100.0	28007285	1,436 k	0	0	0	27645
Ethernet	100.0	27645	1.4	391534	20 k	0	0	0	27645
Internet Protocol Version 4	99.1	27406	2.0	548140	28 k	0	0	0	27406
User Datagram Protocol	78.6	21718	0.6	173744	8,912	0	0	0	21718
Simple Service Discovery Protocol	0.1	26	0.0	11015	565	26	11015	565	26
QUIC IETF	76.9	21269	82.1	22986325	1,179 k	21268	22855316	1,172 k	21563
Malformed Packet	0.0	1	0.0	0	0	1	0	0	1
NetBIOS Name Service	0.0	4	0.0	200	10	4	200	10	4
NetBIOS Datagram Service	0.0	2	0.0	164	8	0	0	0	2
SMB (Server Message Block Protocol)	0.0	2	0.0	238	12	0	0	0	2
SMB MailSlot Protocol	0.0	2	0.0	50	2	0	0	0	2
Microsoft Windows Browser Protocol	0.0	2	0.0	66	3	2	66	3	2
Multicast Domain Name System	0.1	30	0.0	10942	561	30	10942	561	30
Domain Name System	1.4	387	0.1	29611	1,518	387	29611	1,518	387
Transmission Control Protocol	20.6	5683	0.7	203180	10 k	3566	133752	6,860	5683
Transport Layer Security	7.5	2075	9.9	2776846	142 k	2075	2641283	135 k	2082
Hypertext Transfer Protocol	0.1	15	0.0	6640	340	6	2583	132	15
Online Certificate Status Protocol	0.0	4	0.0	1108	56	4	1108	56	4
Line-based text data	0.0	5	0.0	6287	322	5	6287	322	5
Data	0.1	27	0.2	43288	2,220	27	43288	2,220	27
Internet Group Management Protocol	0.0	5	0.0	40	2	5	40	2	5
Address Resolution Protocol	0.9	239	0.0	6692	343	239	6692	343	239

No display filter.

Close Copy Protocols Help

Result and Observations

1. Successfully captured live network traffic using Wireshark.
2. Observed DNS queries and responses during website access.
3. Identified TCP connection establishment using the three-way handshake.
4. Analyzed ICMP packets generated through network diagnostics.
5. Compared HTTP and HTTPS communication.
6. Used protocol hierarchy statistics to understand traffic composition.
7. Applied packet filters to efficiently analyze specific network protocols.

Conclusion

This project provided hands-on experience in capturing and analyzing network traffic using Wireshark. By examining DNS, TCP, ICMP, HTTP, and HTTPS packets, I developed a better understanding of network communication, packet structures, and protocol behavior. The project strengthened my practical knowledge of network monitoring, packet analysis, and troubleshooting, while improving my ability to analyze traffic from a cybersecurity perspective.